```
  ___  ____  ____  ____  ____(R)
 /__    /   ____/   /   ____/
___/   /   /___/   /   /___/
  Statistics/Data analysis
```

```
1 . do "C:\Users\fahim\AppData\Local\Temp\STD962c_000000.tmp"

2 .
3 .
4 .
5 . /* ArterialElastance-with interaction */
6 . destring Elastance, replace force
  Elastance already numeric; no replace

7 .
8 . xtset Number TimeNumerical

  Panel variable: Number (strongly balanced)
   Time variable: TimeNumerical, 0 to 74, but with gaps
           Delta: 1 unit

9 . xtmixed Elastance GroupNumerical##TimeNumerical Sex weight || Number:, var residuals(ar, t(TimeNumerical)) reml
  Note: time gaps exist in the estimation data

  Obtaining starting values by EM:

  Performing gradient-based optimization:

  Iteration 0:   log restricted-likelihood = -60.971441  (not concave)
  Iteration 1:   log restricted-likelihood = -60.912574  (not concave)
  Iteration 2:   log restricted-likelihood = -60.879423  (not concave)
  Iteration 3:   log restricted-likelihood = -60.879358
  Iteration 4:   log restricted-likelihood = -60.879203
  Iteration 5:   log restricted-likelihood = -60.879203  (not concave)
  Iteration 6:   log restricted-likelihood = -60.879203  (not concave)
  Iteration 7:   log restricted-likelihood = -60.879203  (backed up)

  Computing standard errors:

  Mixed-effects REML regression                   Number of obs     =         52
  Group variable: Number                          Number of groups  =         18
                                                  Obs per group:
                                                                min =          2
                                                                avg =        2.9
                                                                max =          3
                                                  Wald chi2(10)     =     107.87
  Log restricted-likelihood = -60.879203          Prob > chi2       =     0.0000
```

Elastance	Coefficient	Std. err.	z	P>|z|	[95% conf.	interval]
GroupNumerical						
1	-.2476525	.4555196	-0.54	0.587	-1.140455	.6451494
2	.4435668	.4611924	0.96	0.336	-.4603537	1.347487
TimeNumerical						
30	-.7418696	.4411383	-1.68	0.093	-1.606485	.1227455
74	-.438074	.4411383	-0.99	0.321	-1.302689	.4265412
GroupNumerical#TimeNumerical						
1 30	.1125811	.6403213	0.18	0.860	-1.142426	1.367588
1 74	2.657973	.6238638	4.26	0.000	1.435222	3.880723
2 30	-.042678	.6238638	-0.07	0.945	-1.265429	1.180073
2 74	2.577615	.6409479	4.02	0.000	1.32138	3.83385
Sex	.0626351	.254634	0.25	0.806	-.4364383	.5617085
weight	-.064503	.0228042	-2.83	0.005	-.1091985	-.0198075

_cons	**6.872465**	**1.805847**	**3.81**	**0.000**	**3.333069**	**10.41186**

Random-effects parameters	Estimate	Std. err.	[95% conf.	interval]
Number: Identity				
var(_cons)	**.035175**	**.1026214**	**.0001156**	**10.70467**
Residual: AR(1)				
rho	**.0013923**	**.**	**.**	**.**
var(e)	**.583809**	**6.06e-49**	**.583809**	**.583809**

LR test vs. linear model: chi2(**2**) = **0.09** Prob > chi2 = **0.9578**

Note: LR test is conservative and provided only for reference.

10 . contrast TimeNumerical@GroupNumerical , effect

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	chi2	P>chi2
Elastance			
TimeNumerical@GroupNumerical			
0	**2**	**2.86**	**0.2394**
1	**2**	**43.46**	**0.0000**
2	**2**	**41.42**	**0.0000**
Joint	**6**	**87.90**	**0.0000**

	Contrast	Std. err.	z	P>|z|	[95% conf.	interval]
Elastance						
TimeNumerical@GroupNumerical						
(30 vs base) 0	**-.7418696**	**.4411383**	**-1.68**	**0.093**	**-1.606485**	**.1227455**
(30 vs base) 1	**-.6292885**	**.4641211**	**-1.36**	**0.175**	**-1.538949**	**.2803721**
(30 vs base) 2	**-.7845476**	**.4411383**	**-1.78**	**0.075**	**-1.649163**	**.0800675**
(74 vs base) 0	**-.438074**	**.4411383**	**-0.99**	**0.321**	**-1.302689**	**.4265412**
(74 vs base) 1	**2.219899**	**.4411383**	**5.03**	**0.000**	**1.355283**	**3.084514**
(74 vs base) 2	**2.139541**	**.4649852**	**4.60**	**0.000**	**1.228187**	**3.050895**

11 . contrast GroupNumerical@TimeNumerical , effect

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	chi2	P>chi2
Elastance			
GroupNumerical@TimeNumerical			
0	**2**	**2.23**	**0.3279**
30	**2**	**1.35**	**0.5099**
74	**2**	**46.19**	**0.0000**
Joint	**6**	**48.41**	**0.0000**

	Contrast	Std. err.	z	P>|z|	[95% conf. interval]	
Elastance						
GroupNumerical@TimeNumerical						
(1 vs base) 0	**-.2476525**	**.4555196**	**-0.54**	**0.587**	**-1.140455**	**.6451494**
(1 vs base) 30	**-.1350714**	**.477127**	**-0.28**	**0.777**	**-1.070223**	**.8000803**
(1 vs base) 74	**2.41032**	**.4555196**	**5.29**	**0.000**	**1.517518**	**3.303122**
(2 vs base) 0	**.4435668**	**.4611924**	**0.96**	**0.336**	**-.4603537**	**1.347487**
(2 vs base) 30	**.4008888**	**.4611924**	**0.87**	**0.385**	**-.5030317**	**1.304809**
(2 vs base) 74	**3.021182**	**.4889579**	**6.18**	**0.000**	**2.062842**	**3.979522**

```
12 .
13 . /* Repeat with natural log */
14 . generate logElastance = ln(Elastance) /*natural log transform*/
   (2 missing values generated)

15 . xtset Number TimeNumerical

   Panel variable: Number (strongly balanced)
    Time variable: TimeNumerical, 0 to 74, but with gaps
            Delta: 1 unit

16 . xtmixed logElastance GroupNumerical##TimeNumerical Sex weight || Number:, var residuals(ar, t(TimeNumerical)) r
   Note: time gaps exist in the estimation data

   Obtaining starting values by EM:

   Performing gradient-based optimization:

   Iteration 0:   log restricted-likelihood = -25.460345  (not concave)
   Iteration 1:   log restricted-likelihood = -25.460207  (backed up)
   Iteration 2:   log restricted-likelihood = -25.460056  (not concave)
   Iteration 3:   log restricted-likelihood = -25.460056  (not concave)
   Iteration 4:   log restricted-likelihood = -25.460056  (not concave)
   Iteration 5:   log restricted-likelihood = -25.460056  (not concave)
   Iteration 6:   log restricted-likelihood = -25.460056  (not concave)
   Iteration 7:   log restricted-likelihood = -25.460056  (not concave)
   Iteration 8:   log restricted-likelihood = -24.734114
   Iteration 9:   log restricted-likelihood = -24.162852
   Iteration 10:  log restricted-likelihood = -24.156119
   Iteration 11:  log restricted-likelihood = -24.156081
   Iteration 12:  log restricted-likelihood = -24.156081

   Computing standard errors:
```

```
Mixed-effects REML regression                   Number of obs     =         52
Group variable: Number                          Number of groups  =         18
                                                Obs per group:
                                                              min =          2
                                                              avg =        2.9
                                                              max =          3
                                                Wald chi2(10)     =     144.05
Log restricted-likelihood = -24.156081          Prob > chi2       =     0.0000
```

logElastance	Coefficient	Std. err.	z	P>|z|	[95% conf.	interval]
GroupNumerical						
1	-.1707086	.2023718	-0.84	0.399	-.56735	.2259328
2	.2135348	.2061158	1.04	0.300	-.1904448	.6175144
TimeNumerical						
30	-.5420356	.1413725	-3.83	0.000	-.8191206	-.2649507
74	-.2987196	.1815917	-1.65	0.100	-.6546328	.0571936
GroupNumerical#TimeNumerical						
1 30	.0361739	.2065811	0.18	0.861	-.3687175	.4410654
1 74	1.073887	.2568095	4.18	0.000	.5705497	1.577224
2 30	.0916816	.1999309	0.46	0.647	-.3001757	.4835389
2 74	1.074102	.2638483	4.07	0.000	.5569692	1.591236
Sex	.0893078	.1384182	0.65	0.519	-.1819869	.3606025
weight	-.036285	.0123669	-2.93	0.003	-.0605236	-.0120463
_cons	3.388613	.9738336	3.48	0.001	1.479934	5.297292

Random-effects parameters	Estimate	Std. err.	[95% conf.	interval]
Number: Identity				
var(_cons)	9.63e-17	2.17e-12	0	.
Residual: AR(1)				
rho	.9776677	.0101331	.9459859	.990854
var(e)	.1218307	.0311923	.0737604	.2012287

LR test vs. linear model: chi2(**2**) = **6.28**   Prob > chi2 = **0.0433**

Note: LR test is conservative and provided only for reference.

17 . contrast TimeNumerical@GroupNumerical , effect

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	chi2	P>chi2
logElastance			
TimeNumerical@GroupNumerical			
0	2	14.92	0.0006
1	2	58.51	0.0000
2	2	53.25	0.0000
Joint	6	126.80	0.0000

	Contrast	Std. err.	z	P>|z|	[95% conf.	interval]
logElastance						
TimeNumerical@GroupNumerical						
(30 vs base) 0	**-.5420356**	**.1413725**	**-3.83**	**0.000**	**-.8191206**	**-.2649507**
(30 vs base) 1	**-.5058617**	**.1506305**	**-3.36**	**0.001**	**-.8010922**	**-.2106313**
(30 vs base) 2	**-.4503541**	**.1413725**	**-3.19**	**0.001**	**-.727439**	**-.1732691**
(74 vs base) 0	**-.2987196**	**.1815917**	**-1.65**	**0.100**	**-.6546328**	**.0571936**
(74 vs base) 1	**.7751674**	**.1815917**	**4.27**	**0.000**	**.4192542**	**1.131081**
(74 vs base) 2	**.7753828**	**.1914168**	**4.05**	**0.000**	**.4002128**	**1.150553**

```
18 . contrast GroupNumerical@TimeNumerical , effect
```

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	chi2	P>chi2
logElastance			
GroupNumerical@TimeNumerical			
0	**2**	**3.33**	**0.1892**
30	**2**	**4.38**	**0.1119**
74	**2**	**39.46**	**0.0000**
Joint	**6**	**45.78**	**0.0000**

	Contrast	Std. err.	z	P>|z|	[95% conf.	interval]
logElastance						
GroupNumerical@TimeNumerical						
(1 vs base) 0	**-.1707086**	**.2023718**	**-0.84**	**0.399**	**-.56735**	**.2259328**
(1 vs base) 30	**-.1345347**	**.2088115**	**-0.64**	**0.519**	**-.5437977**	**.2747283**
(1 vs base) 74	**.9031784**	**.2023718**	**4.46**	**0.000**	**.506537**	**1.29982**
(2 vs base) 0	**.2135348**	**.2061158**	**1.04**	**0.300**	**-.1904448**	**.6175144**
(2 vs base) 30	**.3052164**	**.2061158**	**1.48**	**0.139**	**-.0987632**	**.709196**
(2 vs base) 74	**1.287637**	**.2171067**	**5.93**	**0.000**	**.8621158**	**1.713158**

```
19 .
20 .
21 . /* ArterialElastance -without interaction */
22 . xtset Number TimeNumerical
```

Panel variable: **Number** (strongly balanced)
Time variable: **TimeNumerical, 0** to **74,** but with gaps
Delta: **1 unit**

```
23 . xtmixed Elastance Sex weight i.GroupNumerical i.TimeNumerical || Number:, var reml

   Performing EM optimization:

   Performing gradient-based optimization:

   Iteration 0:   log restricted-likelihood = -74.610358
   Iteration 1:   log restricted-likelihood =  -74.25873
   Iteration 2:   log restricted-likelihood =  -74.25766
   Iteration 3:   log restricted-likelihood =  -74.25766

   Computing standard errors:

   Mixed-effects REML regression                   Number of obs     =         52
   Group variable: Number                          Number of groups  =         18
                                                   Obs per group:
                                                                 min =          2
                                                                 avg =        2.9
                                                                 max =          3
                                                   Wald chi2(6)      =      50.38
   Log restricted-likelihood =  -74.25766          Prob > chi2       =     0.0000
```

Elastance	Coefficient	Std. err.	z	P>|z|	[95% conf.	interval]
Sex	.0634022	.2999618	0.21	0.833	-.5245122	.6513167
weight	-.0619538	.0268735	-2.31	0.021	-.1146249	-.0092827
GroupNumerical						
1	.6890633	.3330347	2.07	0.039	.0363273	1.341799
2	1.249596	.3475872	3.60	0.000	.5683381	1.930855
TimeNumerical						
30	-.7091944	.3310268	-2.14	0.032	-1.357995	-.0603939
74	1.277095	.331225	3.86	0.000	.6279056	1.926284
_cons	6.088209	2.110067	2.89	0.004	1.952553	10.22386

Random-effects parameters	Estimate	Std. err.	[95% conf.	interval]
Number: Identity				
var(_cons)	7.05e-22	4.46e-21	2.92e-27	1.70e-16
var(Residual)	.9565051	.2016492	.6327586	1.445894

```
   LR test vs. linear model: chibar2(01) = 0.00          Prob >= chibar2 = 1.0000

24 .
25 . contrast TimeNumerical@GroupNumerical , effect

   Contrasts of marginal linear predictions

   Margins: asbalanced
```

	df	chi2	P>chi2
Elastance			
TimeNumerical@GroupNumerical			
0	**2**	**35.90**	**0.0000**
1	**2**	**35.90**	**0.0000**
2	**2**	**35.90**	**0.0000**
Joint	**2**	**35.90**	**0.0000**

	Contrast	Std. err.	z	P>|z|	[95% conf. interval]	
Elastance						
TimeNumerical@GroupNumerical						
(30 vs base) 0	**-.7091944**	**.3310268**	**-2.14**	**0.032**	**-1.357995**	**-.0603939**
(30 vs base) 1	**-.7091944**	**.3310268**	**-2.14**	**0.032**	**-1.357995**	**-.0603939**
(30 vs base) 2	**-.7091944**	**.3310268**	**-2.14**	**0.032**	**-1.357995**	**-.0603939**
(74 vs base) 0	**1.277095**	**.331225**	**3.86**	**0.000**	**.6279056**	**1.926284**
(74 vs base) 1	**1.277095**	**.331225**	**3.86**	**0.000**	**.6279056**	**1.926284**
(74 vs base) 2	**1.277095**	**.331225**	**3.86**	**0.000**	**.6279056**	**1.926284**

26 . contrast GroupNumerical@TimeNumerical , effect

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	chi2	P>chi2
Elastance			
GroupNumerical@TimeNumerical			
0	**2**	**13.28**	**0.0013**
30	**2**	**13.28**	**0.0013**
74	**2**	**13.28**	**0.0013**
Joint	**2**	**13.28**	**0.0013**

	Contrast	Std. err.	z	P>|z|	[95% conf. interval]	
Elastance						
GroupNumerical@TimeNumerical						
(1 vs base) 0	**.6890633**	**.3330347**	**2.07**	**0.039**	**.0363273**	**1.341799**
(1 vs base) 30	**.6890633**	**.3330347**	**2.07**	**0.039**	**.0363273**	**1.341799**
(1 vs base) 74	**.6890633**	**.3330347**	**2.07**	**0.039**	**.0363273**	**1.341799**
(2 vs base) 0	**1.249596**	**.3475872**	**3.60**	**0.000**	**.5683381**	**1.930855**
(2 vs base) 30	**1.249596**	**.3475872**	**3.60**	**0.000**	**.5683381**	**1.930855**
(2 vs base) 74	**1.249596**	**.3475872**	**3.60**	**0.000**	**.5683381**	**1.930855**

```
27 .
28 .
29 . /* EjectionFraction-with interaction  */
30 . destring EjectionFraction, replace force
   EjectionFraction already numeric; no replace

31 .
32 . xtset Number TimeNumerical

   Panel variable: Number (strongly balanced)
    Time variable: TimeNumerical, 0 to 74, but with gaps
            Delta: 1 unit

33 . xtmixed EjectionFraction GroupNumerical##TimeNumerical Sex weight || Number:, var reml

   Performing EM optimization:

   Performing gradient-based optimization:

   Iteration 0:   log restricted-likelihood = -161.15832
   Iteration 1:   log restricted-likelihood =  -161.1582
   Iteration 2:   log restricted-likelihood =  -161.1582

   Computing standard errors:

   Mixed-effects REML regression                   Number of obs     =         52
   Group variable: Number                          Number of groups  =         18
                                                   Obs per group:
                                                                 min =          2
                                                                 avg =        2.9
                                                                 max =          3
                                                   Wald chi2(10)     =      60.50
   Log restricted-likelihood =  -161.1582          Prob > chi2       =     0.0000
```

EjectionFraction	Coefficient	Std. err.	z	P>|z|	[95% conf.	interval]
GroupNumerical						
1	.5014744	5.624076	0.09	0.929	-10.52151	11.52446
2	.2652198	5.728773	0.05	0.963	-10.96297	11.49341
TimeNumerical						
30	16.81036	4.494287	3.74	0.000	8.001722	25.619
74	11.57262	4.494287	2.57	0.010	2.76398	20.38126
GroupNumerical#TimeNumerical						
1 30	.3327281	6.554893	0.05	0.960	-12.51463	13.18008
1 74	-8.249626	6.355881	-1.30	0.194	-20.70692	4.207672
2 30	.8787463	6.355881	0.14	0.890	-11.57855	13.33604
2 74	-17.61846	6.558819	-2.69	0.007	-30.47351	-4.763412
Sex	.8398223	3.886476	0.22	0.829	-6.777532	8.457176
weight	.5735463	.3447634	1.66	0.096	-.1021774	1.24927
_cons	-3.394594	27.14134	-0.13	0.900	-56.59065	49.80146

Random-effects parameters	Estimate	Std. err.	[95% conf.	interval]
Number: Identity				
var(_cons)	**33.49252**	**22.13566**	**9.170168**	**122.3259**
var(Residual)	**60.59584**	**16.15711**	**35.93195**	**102.1892**

LR test vs. linear model: chibar2(01) = **4.44**   Prob >= chibar2 = **0.0176**

34 . contrast TimeNumerical@GroupNumerical , effect

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	chi2	P>chi2
EjectionFraction			
TimeNumerical@GroupNumerical			
0	**2**	**14.65**	**0.0007**
1	**2**	**14.07**	**0.0009**
2	**2**	**27.97**	**0.0000**
Joint	**6**	**56.74**	**0.0000**

	Contrast	Std. err.	z	P>|z|	[95% conf.	interval]
EjectionFraction						
TimeNumerical@GroupNumerical						
(30 vs base) 0	**16.81036**	**4.494287**	**3.74**	**0.000**	**8.001722**	**25.619**
(30 vs base) 1	**17.14309**	**4.771583**	**3.59**	**0.000**	**7.790959**	**26.49522**
(30 vs base) 2	**17.68911**	**4.494287**	**3.94**	**0.000**	**8.880468**	**26.49775**
(74 vs base) 0	**11.57262**	**4.494287**	**2.57**	**0.010**	**2.76398**	**20.38126**
(74 vs base) 1	**3.322994**	**4.494287**	**0.74**	**0.460**	**-5.485646**	**12.13163**
(74 vs base) 2	**-6.045841**	**4.776976**	**-1.27**	**0.206**	**-15.40854**	**3.316859**

35 . contrast GroupNumerical@TimeNumerical , effect

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	chi2	P>chi2
EjectionFraction			
GroupNumerical@TimeNumerical			
0	**2**	**0.01**	**0.9960**
30	**2**	**0.04**	**0.9786**
74	**2**	**8.37**	**0.0152**
Joint	**6**	**11.04**	**0.0873**

	Contrast	Std. err.	z	P>|z|	[95% conf.	interval]
EjectionFraction						
GroupNumerical@TimeNumerical						
(1 vs base) 0	**.5014744**	**5.624076**	**0.09**	**0.929**	**-10.52151**	**11.52446**
(1 vs base) 30	**.8342025**	**5.841377**	**0.14**	**0.886**	**-10.61469**	**12.28309**
(1 vs base) 74	**-7.748152**	**5.624076**	**-1.38**	**0.168**	**-18.77114**	**3.274834**
(2 vs base) 0	**.2652198**	**5.728773**	**0.05**	**0.963**	**-10.96297**	**11.49341**
(2 vs base) 30	**1.143966**	**5.728773**	**0.20**	**0.842**	**-10.08422**	**12.37216**
(2 vs base) 74	**-17.35324**	**6.001395**	**-2.89**	**0.004**	**-29.11576**	**-5.590723**

```
36 .
37 .
38 . /* EjectionFraction -without interaction */
39 . xtset Number TimeNumerical

   Panel variable: Number (strongly balanced)
    Time variable: TimeNumerical, 0 to 74, but with gaps
            Delta: 1 unit

40 . xtmixed EjectionFraction Sex weight i.GroupNumerical i.TimeNumerical || Number:, var reml

   Performing EM optimization:

   Performing gradient-based optimization:

   Iteration 0:   log restricted-likelihood = -176.51488
   Iteration 1:   log restricted-likelihood = -176.51453
   Iteration 2:   log restricted-likelihood = -176.51453

   Computing standard errors:

   Mixed-effects REML regression                  Number of obs     =         52
   Group variable: Number                         Number of groups  =         18
                                                  Obs per group:
                                                                min =          2
                                                                avg =        2.9
                                                                max =          3
                                                  Wald chi2(6)      =      43.82
   Log restricted-likelihood = -176.51453         Prob > chi2       =     0.0000
```

EjectionFraction	Coefficient	Std. err.	z	P>|z|	[95% conf.	interval]
Sex	**.7606436**	**3.952766**	**0.19**	**0.847**	**-6.986636**	**8.507923**
weight	**.529861**	**.3505888**	**1.51**	**0.131**	**-.1572803**	**1.217002**
GroupNumerical						
1	**-2.172169**	**4.365843**	**-0.50**	**0.619**	**-10.72906**	**6.384726**
2	**-4.706639**	**4.529616**	**-1.04**	**0.299**	**-13.58452**	**4.171245**
TimeNumerical						
30	**17.24504**	**2.85883**	**6.03**	**0.000**	**11.64183**	**22.84824**
74	**3.405072**	**2.859861**	**1.19**	**0.234**	**-2.200153**	**9.010296**
_cons	**2.660563**	**27.48659**	**0.10**	**0.923**	**-51.21216**	**56.53329**

Random-effects parameters	Estimate	Std. err.	[95% conf.	interval]
Number: Identity				
var(_cons)	**31.90751**	**23.1267**	**7.707942**	**132.0832**
var(Residual)	**70.90258**	**17.7249**	**43.43807**	**115.732**

LR test vs. linear model: chibar2(01) = **3.50**   Prob >= chibar2 = **0.0307**

41 .
42 . contrast TimeNumerical@GroupNumerical , effect

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	chi2	P>chi2
EjectionFraction			
TimeNumerical@GroupNumerical			
0	**2**	**40.21**	**0.0000**
1	**2**	**40.21**	**0.0000**
2	**2**	**40.21**	**0.0000**
Joint	**2**	**40.21**	**0.0000**

	Contrast	Std. err.	z	P>|z|	[95% conf.	interval]
EjectionFraction						
TimeNumerical@GroupNumerical						
(30 vs base) 0	**17.24504**	**2.85883**	**6.03**	**0.000**	**11.64183**	**22.84824**
(30 vs base) 1	**17.24504**	**2.85883**	**6.03**	**0.000**	**11.64183**	**22.84824**
(30 vs base) 2	**17.24504**	**2.85883**	**6.03**	**0.000**	**11.64183**	**22.84824**
(74 vs base) 0	**3.405072**	**2.859861**	**1.19**	**0.234**	**-2.200153**	**9.010296**
(74 vs base) 1	**3.405072**	**2.859861**	**1.19**	**0.234**	**-2.200153**	**9.010296**
(74 vs base) 2	**3.405072**	**2.859861**	**1.19**	**0.234**	**-2.200153**	**9.010296**

43 . contrast GroupNumerical@TimeNumerical , effect

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	chi2	P>chi2
EjectionFraction			
GroupNumerical@TimeNumerical			
0	**2**	**1.08**	**0.5821**
30	**2**	**1.08**	**0.5821**
74	**2**	**1.08**	**0.5821**
Joint	**2**	**1.08**	**0.5821**

	Contrast	Std. err.	z	P>|z|	[95% conf.	interval]
EjectionFraction						
GroupNumerical@TimeNumerical						
(1 vs base) 0	**-2.172169**	**4.365843**	**-0.50**	**0.619**	**-10.72906**	**6.384726**
(1 vs base) 30	**-2.172169**	**4.365843**	**-0.50**	**0.619**	**-10.72906**	**6.384726**
(1 vs base) 74	**-2.172169**	**4.365843**	**-0.50**	**0.619**	**-10.72906**	**6.384726**
(2 vs base) 0	**-4.706639**	**4.529616**	**-1.04**	**0.299**	**-13.58452**	**4.171245**
(2 vs base) 30	**-4.706639**	**4.529616**	**-1.04**	**0.299**	**-13.58452**	**4.171245**
(2 vs base) 74	**-4.706639**	**4.529616**	**-1.04**	**0.299**	**-13.58452**	**4.171245**

```
44 .
45 .
   end of do-file

46 .
```